

```
(%i4) /* Symbols */
declare(h, [real, constant]);
assume(h > 0);
declare(C, [real, constant]);
depends(I, h);

→ romberg(steps) := block( /* Romberg rule: multiple Richardson extrapolations */
    [local_I1, local_I2, C_val],
    local_I1: I(h) + C·h^2, /* Start with a low order (p = 2) method */
    local_I2: subst(h = h / 2, local_I1),

    for i: 1 thru steps do (
        C_val: rhs(solve(local_I1 = local_I2, C))[1],
        local_I1: factor(subst(C = C_val, local_I1)) + C · h^(2 · i + 2),
        local_I2: subst(h = h / 2, local_I1)
    ),
    return(local_I1)
);$;
```

```
(%i6) for i: 1 thru 7 do ( /* Integral estimations */
    print(romberg(i))
);$;
```

$$C h^4 - \frac{I(h) - 4 I\left(\frac{h}{2}\right)}{3}$$

$$\frac{I(h) - 20 I\left(\frac{h}{2}\right) + 64 I\left(\frac{h}{4}\right)}{45} + C h^6$$

$$C h^8 - \frac{I(h) - 84 I\left(\frac{h}{2}\right) + 1344 I\left(\frac{h}{4}\right) - 4096 I\left(\frac{h}{8}\right)}{2835}$$

$$\frac{I(h) - 340 I\left(\frac{h}{2}\right) + 22848 I\left(\frac{h}{4}\right) - 348160 I\left(\frac{h}{8}\right) + 1048576 I\left(\frac{h}{16}\right)}{722925} + C h^{10}$$

$$C h^{12} - \frac{I(h) - 1364 I\left(\frac{h}{2}\right) + 371008 I\left(\frac{h}{4}\right) - 23744512 I\left(\frac{h}{8}\right) + 357564416 I\left(\frac{h}{16}\right) - 1073741824 I\left(\frac{h}{32}\right)}{739552275}$$

$$I(h) - 5460 I\left(\frac{h}{2}\right) + 5957952 I\left(\frac{h}{4}\right) - 1543393280 I\left(\frac{h}{8}\right) + 97615085568 I\left(\frac{h}{16}\right) - 1465657589760$$

$$I\left(\frac{h}{32}\right) + 4398046511104 I\left(\frac{h}{64}\right) / 3028466566125 + C h^{14}$$

$$C h^{16} - I(h) - 21844 I\left(\frac{h}{2}\right) + 95414592 I\left(\frac{h}{4}\right) - 99158478848 I\left(\frac{h}{8}\right) + 25384570585088 I\left(\frac{h}{16}\right) -$$

$$1600791219535872 I\left(\frac{h}{32}\right) + 24017731997138944 I\left(\frac{h}{64}\right) - 72057594037927936 I\left(\frac{h}{128}\right) /$$

$$49615367752825875$$